■ CHVN Paper Reviewer

Multi-Agent Scientific Paper Review

qwen3:8b · May 04, 2026 07:13

File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	DEEP
Model	qwen3:8b
Target Publisher	IEEE

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Editorial Decision

MAJOR REVISION

Weighted Score: 3.5 / 5.0
Confidence: 84%

Executive Summary

The manuscript presents a novel physics-informed EANN framework for structural parameter identification under multi-objective optimization. While the methodology and experimental validation are well-developed, several critical issues must be addressed to meet IEEE standards for technical rigor and reproducibility.

Decision Rationale

The manuscript demonstrates significant technical contribution and experimental validation, but it requires major revisions to address critical issues in reproducibility, literature review, and clarity of methodology. These revisions are essential to meet IEEE standards for technical rigor, mathematical correctness, and experimental validation.

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**5.0**	Excellent	■■■■■
Manuscript Structure	**2.0**	Weak	■■■■■
Methodology & Equations	**4.0**	Good	■■■■■
Statistical Analysis	**4.5**	Excellent	■■■■■
Results & Evidence	**4.2**	Good	■■■■■

Discussion & Conclusions	**3.5**	Good	■■■■■
Figures & Tables	**1.5**	Very Weak	■■■■■
Scientific Writing	**3.0**	Acceptable	■■■■■
References & Citations	**2.5**	Weak	■■■■■
Ethics & Transparency	**1.0**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **5.0 / 5.0** | Confidence: 100%

Strengths:

- [TITLE] The title is specific, informative, and includes key terms for discoverability.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions.
- [ABSTRACT] The abstract quantifies results with statistical measures and avoids vague language.
- [KEYWORDS] The keywords are comprehensive, include domain-specific terms, and are not redundant with the title.

Minor Comments:

- [ABSTRACT] The abstract could be slightly more concise, but it is not a critical issue.

Recommendations:

- [TITLE] Consider shortening the title slightly for improved readability, though it is already within acceptable length.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.0 / 5.0** | Confidence: 95%

Strengths:

- Clear title and keywords
- Reproducibility of methodology is emphasized
- Experimental validation is described

Weaknesses:

- Missing literature review section
- Research problem not clearly stated in introduction
- Research question or hypothesis not explicitly declared
- Objectives not stated or addressed in conclusions
- Section VI and VII are duplicated and redundant
- Section transitions are unclear
- Section balance is uneven with excessive repetition
- Consistency between objectives, methods, results, and conclusions is lacking
- Abstract claims do not match the results
- Keywords are not aligned with content

Major Comments:

- The manuscript is missing a critical literature review section, which is essential for establishing the context and novelty of the work.
- The research problem is not clearly stated in the introduction, making it difficult to assess the significance of the study.
- The research question or hypothesis is not explicitly declared, which undermines the clarity and focus of the study.
- The objectives are not stated or addressed in the conclusions, leading to a disconnect between the stated goals and the final outcomes.
- Sections VI and VII are duplicated and redundant, which compromises the logical flow and coherence of the manuscript.

- The transitions between sections are unclear, making it difficult to follow the logical progression of the work.
- The section balance is uneven, with excessive repetition and an absence of key sections such as the literature review.
- The consistency between objectives, methods, results, and conclusions is lacking, which affects the overall coherence of the manuscript.
- The abstract claims do not match the results, which raises concerns about the accuracy and completeness of the summary.
- The keywords are not aligned with the content, which may affect the discoverability and relevance of the manuscript.

Minor Comments:

- The title is somewhat vague and could be more specific to reflect the content of the study.
- The methodology section is incomplete and lacks sufficient detail for reproducibility.
- The experimental results are not fully described, and the figures are not properly labeled or referenced.
- The discussion section is incomplete and does not adequately interpret the results.
- The references are not properly formatted and lack recent citations.

Recommendations:

- Add a comprehensive literature review section to establish the context and novelty of the work.
- Clearly state the research problem in the introduction to highlight the significance of the study.
- Explicitly declare the research question or hypothesis to provide a clear focus for the study.
- State the objectives clearly and ensure they are addressed in the conclusions.
- Remove the duplicated sections VI and VII to improve the logical flow and coherence of the manuscript.
- Improve the transitions between sections to enhance the logical progression of the work.
- Ensure the section balance is appropriate, with all key sections included and properly developed.
- Ensure consistency between objectives, methods, results, and conclusions to improve the overall coherence of the manuscript.
- Revise the abstract to accurately reflect the results and ensure it matches the content of the manuscript.
- Align the keywords with the content of the manuscript to improve discoverability and relevance.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 95%

Strengths:

- The paper presents a novel physics-informed EANN framework for structural parameter identification under multi-objective optimization.
- The experimental validation is thorough, with statistical analysis and comparison against GA and PSO baselines.
- The paper addresses the practical challenge of acceleration-only identification, which is relevant to real-world seismic monitoring applications.

Weaknesses:

- The methodology section is poorly organized and lacks a clear, logical flow, which may hinder reproducibility and understanding.
- The mathematical formulation is not fully explained, and some equations are not properly referenced or defined.
- The paper does not provide sufficient detail on the implementation of the EANN framework, which may limit reproducibility.

Major Comments:

- The methodology section is fragmented and lacks a clear, cohesive structure. The authors should reorganize the methodology to ensure a logical flow from problem definition to solution implementation.

- The mathematical formulation is not fully explained, and some equations are not properly referenced or defined. This may hinder reproducibility and understanding.

Minor Comments:

- The paper should provide more detail on the implementation of the EANN framework to ensure reproducibility.
- Some equations are not consistently defined or referenced in the text.

Recommendations:

- Reorganize the methodology section to ensure a clear, logical flow from problem definition to solution implementation.
- Provide a more detailed explanation of the mathematical formulation and ensure that all equations are properly referenced and defined.
- Include additional implementation details of the EANN framework to improve reproducibility.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.5 / 5.0** | Confidence: 95%

Strengths:

- Use of nonparametric tests (Kruskal-Wallis and Wilcoxon) is appropriate given the non-normal distribution of the data.
- Bootstrap confidence intervals are used to quantify uncertainty in parameter estimates.
- Coefficient of variation (CV) is used to assess parametric robustness across runs.
- Hypervolume and Pareto front density are used to assess multi-objective trade-offs.
- Smoothness index is used to quantify the spatial continuity of identified parameters.
- The manuscript reports 30 independent runs, which is sufficient for assessing robustness in stochastic optimization.
- The manuscript reports RMSE, MAE, and R^2 as primary performance metrics, which are appropriate for evaluating model accuracy.
- The manuscript reports computational cost per stochastic run, which is important for assessing algorithm efficiency.

Weaknesses:

- Bootstrap confidence intervals are reported but not explicitly tied to specific metrics or parameters, making it difficult to assess their relevance to the analysis.
- The manuscript does not report the number of bootstrap resamples used, which is critical for assessing the reliability of bootstrap confidence intervals.
- The manuscript does not report the exact p-values for the Wilcoxon rank-sum tests, only that they are < 0.001, which is insufficient for detailed interpretation.
- The manuscript does not report the exact p-values for the Kruskal-Wallis test, only that they are < 0.001, which is insufficient for detailed interpretation.
- The manuscript does not report the exact p-values for the post-hoc Wilcoxon tests, only that they are < 0.001, which is insufficient for detailed interpretation.

Major Comments:

- The manuscript should report the exact p-values for all statistical tests, not just the conclusion that they are < 0.001.
- The manuscript should report the number of bootstrap resamples used to construct confidence intervals, as this is critical for assessing their reliability.
- The manuscript should clarify how bootstrap confidence intervals are used in the analysis, particularly in relation to specific metrics or parameters.

Minor Comments:

- The manuscript should report the exact p-values for the Wilcoxon rank-sum tests, not just that they are < 0.001.

- The manuscript should report the exact p-values for the Kruskal-Wallis test, not just that they are < 0.001.
- The manuscript should report the exact p-values for the post-hoc Wilcoxon tests, not just that they are < 0.001.

Recommendations:

- Report the exact p-values for all statistical tests, including the Kruskal-Wallis and Wilcoxon rank-sum tests.
- Report the number of bootstrap resamples used to construct confidence intervals.
- Clarify how bootstrap confidence intervals are used in the analysis, particularly in relation to specific metrics or parameters.
- Report the exact p-values for all statistical tests, including the Kruskal-Wallis and Wilcoxon rank-sum tests.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **1.5 / 5.0** | Confidence: 50%

Strengths:

- The manuscript presents a clear and structured approach to structural parameter identification using a multi-objective optimization framework.
- The methodology is well-defined and includes a state-space formulation and modal decomposition, which are standard in structural dynamics.
- The text provides a detailed description of the experimental setup, including the use of accelerometers and displacement sensors.
- The manuscript includes a comprehensive discussion of the results, including statistical inference and comparison with existing algorithms.
- The manuscript emphasizes the importance of physical plausibility and robustness in structural identification.

Weaknesses:

- The figures and tables referenced in the text are not provided, making it impossible to verify the data or its consistency with the text.
- The equations referenced in the text are not provided, making it impossible to verify their content or their consistency with the text.
- The text makes claims about the results and statistical significance, but the supporting figures and tables are not available for verification.
- The manuscript lacks detailed descriptions of the actual figures and tables, which reduces the self-explanatory nature of the content.
- The manuscript does not provide the actual data or visualizations that support the claims made in the text, which is critical for reproducibility and validation.

Major Comments:

- The manuscript fails to provide the actual figures, tables, and equations referenced in the text. This is a critical issue for reproducibility and validation.
- The text makes claims about the results and statistical significance, but the supporting data is not available, which undermines the credibility of the findings.
- The manuscript lacks detailed descriptions of the actual figures and tables, which reduces the self-explanatory nature of the content.
- The manuscript does not provide the actual data or visualizations that support the claims made in the text, which is critical for reproducibility and validation.

Minor Comments:

- The text could benefit from more detailed descriptions of the figures and tables to improve clarity and self-explanatory nature.
- The equations referenced in the text should be provided to ensure mathematical correctness and consistency.
- The manuscript could benefit from more detailed descriptions of the experimental setup and data preprocessing steps to improve reproducibility.

Recommendations:

- Provide the actual figures, tables, and equations referenced in the text to ensure reproducibility and validation.
- Include detailed descriptions of the figures and tables to improve clarity and self-explanatory nature.
- Provide the actual equations referenced in the text to ensure mathematical correctness and consistency.
- Include more detailed descriptions of the experimental setup and data preprocessing steps to improve reproducibility.

ResultsReviewer

Results validity and empirical support

Score: **4.2 / 5.0** | Confidence: 85%

Strengths:

- The results clearly address the stated objectives of the study, including the comparison of the EANN framework against GA and PSO baselines.
- Quantitative results are well-supported by MSE, RMSE, CV, and Pareto analysis, which are directly tied to the experimental setup and validation.
- The manuscript provides a comprehensive statistical validation using Kruskal-Wallis, Wilcoxon rank-sum tests, and nonparametric bootstrap confidence intervals.
- The results are consistent across text, figures, and tables, with clear reporting of median RMSE values and their comparison to the PSO baseline.
- The manuscript includes a detailed description of the experimental setup, preprocessing steps, and evaluation protocol, which supports reproducibility.

Weaknesses:

- Negative results are not reported. The manuscript does not mention any cases where the EANN framework underperformed or failed to meet expectations, which is critical for a complete empirical evaluation.
- The Pareto front analysis is mentioned but not fully detailed in the results section. The manuscript should clarify how the Pareto front was generated and interpreted, including the specific metrics used for dominance.
- The hypervolume and spread indicators are referenced but not explicitly reported in the results. This omission reduces the completeness of the multi-objective optimization analysis.
- The CV (coefficient of variation) is mentioned but not fully contextualized. The manuscript should clarify how the CV values (CV ≈0.07-0.10) were calculated and what they imply for parametric dispersion.
- The results section does not explicitly state whether the 30 independent runs were conducted with identical experimental conditions or if there were any variations in the input data or parameters across runs.

Major Comments:

- The manuscript lacks a discussion of negative results, which is essential for a balanced and rigorous evaluation of the proposed framework. This omission could undermine the credibility of the results and the claims made in the abstract and conclusion.
- The Pareto front analysis and hypervolume/spread indicators are not sufficiently detailed in the results section. The manuscript should provide a more comprehensive description of the multi-objective optimization outcomes, including how dominance was determined and how the front was visualized or quantified.

Minor Comments:

- The CV values are mentioned but not fully explained in the context of the results. The manuscript should clarify how these values were calculated and their significance in the context of parametric dispersion.
- The results section does not explicitly state whether the 30 independent runs were conducted under identical experimental conditions or if there were any variations in the input data or parameters across runs. This information is critical for assessing the robustness and reproducibility of the results.
- The results section should explicitly reference the figures and tables that contain the quantitative results (e.g., MSE, RMSE, CV, and Pareto front data) to ensure clarity and completeness.

Recommendations:

• Include a section in the results discussing negative results or cases where the EANN framework underperformed, to provide a more balanced evaluation.

• Clarify the methodology for generating and interpreting the Pareto front, including the specific metrics used for dominance and how the front was visualized or quantified.

• Provide detailed reporting of the hypervolume and spread indicators, including their calculation and interpretation in the context of the multi-objective optimization problem.

• Elaborate on the CV values, including their calculation and how they relate to the parametric dispersion in the damping coefficients.

• Ensure that all quantitative results (e.g., MSE, RMSE, CV, and Pareto front data) are explicitly tied to the corresponding figures and tables, and that the results section clearly references these visual elements.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 90%

Strengths:

• The Discussion section effectively contextualizes the results within the framework of structural identification and control systems, highlighting the advantages of EANN over GA and PSO.

• The Conclusions section clearly states the contribution of the paper and aligns with the experimental results.

• The manuscript provides a clear statement of the problem and its significance, which is consistent with the introduction and abstract.

Weaknesses:

• The Discussion section lacks a comprehensive comparison with prior work in the field of structural identification and control systems. It only cites references but does not engage in a detailed comparison of methodologies or results.

• The Discussion does not address unexpected findings or anomalies in the results, such as the occasional lower median error values achieved by GA.

• The manuscript does not clearly state the limitations of the study, such as the computational cost of EANN or the assumption of a five-story building prototype.

• The Discussion section does not discuss the theoretical or practical implications of the findings in sufficient depth.

• The Conclusions section does not provide specific, quantified statements about the contributions of the work, such as the exact improvement in accuracy or efficiency over existing methods.

• The Conclusions section does not specify future work directions, which are necessary for guiding further research.

Major Comments:

• The Discussion section fails to engage with the existing literature, which is critical for establishing the novelty and significance of the proposed EANN framework.

• The manuscript does not adequately address the limitations of the study, such as computational cost and generalizability beyond the five-story building prototype.

• The Conclusions section lacks specificity and quantification, which is essential for demonstrating the impact of the work.

Minor Comments:

• The Discussion section should clarify the theoretical implications of the results, such as how the EANN framework contributes to the field of structural health monitoring.

• The manuscript should provide more context on the experimental setup and assumptions, particularly regarding the five-story building prototype.

Recommendations:

• Include a detailed comparison with prior work in the Discussion section, highlighting how the EANN framework improves upon existing methods.

- Clarify the limitations of the study, such as computational cost and the scope of the experimental setup.
- Provide specific, quantified statements in the Conclusions section to emphasize the contributions of the work.
- Specify future work directions, such as extending the framework to larger structures or integrating it with real-time monitoring systems.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.0 / 5.0** | Confidence: 55%

Strengths:

- The manuscript contains enough extracted text for a representative writing review.
- Technical terminology and quantitative reporting signals are present in the extracted text.

Weaknesses:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Minor Comments:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Recommendations:

- Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
- Ensure technical terms are defined on first use and used consistently across sections.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- References are numbered and formatted consistently
- Recent references from 2021-2023 are included
- Citations are relevant to the topic of structural parameter identification

Weaknesses:

- Missing seminal foundational works in structural dynamics and control theory
- Lack of key references on structural health monitoring and observer-based state reconstruction
- Missing references on seismic design regulations and their relation to structural models

Major Comments:

- The references section lacks seminal foundational works in structural dynamics and control theory, which are critical for establishing the theoretical basis of the work.
- Key areas such as structural health monitoring and observer-based state reconstruction are underrepresented in the references, which weakens the literature review and contextualization of the work.
- The manuscript does not include references to seminal works on seismic design regulations, which are directly mentioned in the introduction and are essential for understanding the practical relevance of the models.

Minor Comments:

- The reference [29] is cited as a book but lacks a publication year, which is inconsistent with the citation style used for other references.
- The reference [34] is cut off and incomplete, which may affect the reproducibility and completeness of the literature review.

Recommendations:

- Include seminal foundational works in structural dynamics and control theory, such as those by Chopra, K. S. Narendra, and others.
- Add references to key works on structural health monitoring and observer-based state reconstruction to strengthen the literature review.
- Include references to seismic design regulations and their relation to structural models, such as those from Mexico City, to contextualize the practical relevance of the work.
- Ensure all references are complete and properly formatted, including publication years for books.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.0 / 5.0** | Confidence: 95%

Strengths:

- The methodology section provides a detailed formulation of the mathematical framework for structural identification.
- The experimental results are presented with figures and tables that support the claims made in the discussion and conclusions.
- The paper presents a novel framework for structural parameter identification using a physics-informed EANN approach.

Weaknesses:

- The manuscript lacks ethical approval information, which is critical for studies involving human or animal subjects.
- The data availability statement is missing, which is essential for reproducibility and transparency.
- Conflict of interest is not declared, which is a key requirement for ethical research.
- Funding sources are not disclosed, which is necessary for transparency.
- Author contributions are not stated, which is important for proper attribution.
- Limitations are not explicitly stated, which is a significant omission in the discussion section.
- The limitations are not specific or detailed, which undermines the credibility of the conclusions.
- The impact of limitations on the conclusions is not assessed, which is a critical aspect of limitations reporting.
- The generalizability of findings is not appropriately qualified, which is important for interpreting the results.
- Future work does not address current limitations, which is a key aspect of limitations reporting.
- Reproducibility is not addressed, as data and code availability are not provided, which is a fundamental requirement for scientific validation.

Major Comments:

- The manuscript fails to report ethical approval, which is a critical requirement for any research involving human or animal subjects. This omission raises serious ethical concerns.
- The absence of a data availability statement undermines the reproducibility and transparency of the research, which is essential for scientific integrity.
- Conflict of interest is not declared, which is a fundamental requirement for ethical research and trust in the findings.
- Funding sources are not disclosed, which is necessary for transparency and accountability in research.
- Author contributions are not stated, which is important for proper attribution and accountability.
- Limitations are not explicitly stated, which is a significant omission in the discussion section and undermines the credibility of the conclusions.
- The limitations are not specific or detailed, which is a critical aspect of limitations reporting.
- The impact of limitations on the conclusions is not assessed, which is important for interpreting the results.
- The generalizability of findings is not appropriately qualified, which is important for interpreting the results.
- Future work does not address current limitations, which is a key aspect of limitations reporting.

• Reproducibility is not addressed, as data and code availability are not provided, which is a fundamental requirement for scientific validation.

Minor Comments:

• The methodology section is detailed, but the lack of ethical and transparency information is a major issue.

• The discussion section presents results but lacks a thorough analysis of limitations.

Recommendations:

• The manuscript should include a statement of ethical approval for any human or animal studies.

• A data availability statement should be included to ensure transparency and reproducibility.

• Conflict of interest should be declared to maintain the integrity of the research.

• Funding sources should be disclosed to ensure transparency.

• Author contributions should be stated to ensure proper attribution.

• Limitations should be explicitly stated and discussed in detail in the discussion section.

• The impact of limitations on the conclusions should be assessed and discussed.

• The generalizability of findings should be appropriately qualified.

• Future work should address the current limitations to improve the research.

• Reproducibility should be addressed by providing data and code availability.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] The manuscript is missing a critical literature review section, which is essential for establishing the context and novelty of the work.

[2] The research problem is not clearly stated in the introduction, making it difficult to assess the significance of the study.

[3] The research question or hypothesis is not explicitly declared, which undermines the clarity and focus of the study.

[4] The objectives are not stated or addressed in the conclusions, leading to a disconnect between the stated goals and the final outcomes.

[5] Sections VI and VII are duplicated and redundant, which compromises the logical flow and coherence of the manuscript.

[6] The transitions between sections are unclear, making it difficult to follow the logical progression of the work.

[7] The section balance is uneven, with excessive repetition and an absence of key sections such as the literature review.

[8] The consistency between objectives, methods, results, and conclusions is lacking, which affects the overall coherence of the manuscript.

[9] The abstract claims do not match the results, which raises concerns about the accuracy and completeness of the summary.

[10] The keywords are not aligned with the content, which may affect the discoverability and relevance of the manuscript.

[11] The methodology section is fragmented and lacks a clear, cohesive structure. The authors should reorganize the methodology to ensure a logical flow from problem definition to solution implementation.

[12] The mathematical formulation is not fully explained, and some equations are not properly referenced or defined. This may hinder reproducibility and understanding.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] [ABSTRACT] The abstract could be slightly more concise, but it is not a critical issue.

[2] The title is somewhat vague and could be more specific to reflect the content of the study.

[3] The methodology section is incomplete and lacks sufficient detail for reproducibility.

[4] The experimental results are not fully described, and the figures are not properly labeled or referenced.

[5] The discussion section is incomplete and does not adequately interpret the results.

[6] The references are not properly formatted and lack recent citations.

[7] The paper should provide more detail on the implementation of the EANN framework to ensure reproducibility.

[8] Some equations are not consistently defined or referenced in the text.

[9] The manuscript should report the exact p-values for the Wilcoxon rank-sum tests, not just that they are < 0.001.

[10] The manuscript should report the exact p-values for the Kruskal-Wallis test, not just that they are < 0.001.

[11] The manuscript should report the exact p-values for the post-hoc Wilcoxon tests, not just that they are $<$ 0.001.

[12] The text could benefit from more detailed descriptions of the figures and tables to improve clarity and self-explanatory nature.

Specific Improvement Recommendations

1. [TITLE] Consider shortening the title slightly for improved readability, though it is already within acceptable length.
2. Add a comprehensive literature review section to establish the context and novelty of the work.
3. Clearly state the research problem in the introduction to highlight the significance of the study.
4. Explicitly declare the research question or hypothesis to provide a clear focus for the study.
5. State the objectives clearly and ensure they are addressed in the conclusions.
6. Remove the duplicated sections VI and VII to improve the logical flow and coherence of the manuscript.
7. Improve the transitions between sections to enhance the logical progression of the work.
8. Ensure the section balance is appropriate, with all key sections included and properly developed.
9. Ensure consistency between objectives, methods, results, and conclusions to improve the overall coherence of the manuscript.
10. Revise the abstract to accurately reflect the results and ensure it matches the content of the manuscript.
11. Align the keywords with the content of the manuscript to improve discoverability and relevance.
12. Reorganize the methodology section to ensure a clear, logical flow from problem definition to solution implementation.
